RS-vHeat: Heat Conduction Guided Efficient Remote Sensing Foundation Model

Huiyang Hu[1,2,3,4] Peijin Wang[1,2,3,4] * Hanbo Bi[1,2,3,4] Boyuan Tong[1,2,3,4] Zhaozhi Wang[2,3,5]
Wenhui Diao[1,2,3,4] Hao Chang[1,2,3,4] Yingchao Feng[1,2,3,4] Ziqi Zhang[2]
Yaowei Wang[5] Qixiang Ye[2,3,5] Kun Fu[1,2,3,4] Xian Sun[1,2,3,4]
[1]Aerospace Information Research Institute, Chinese Academy of Sciences
[2]University of Chinese Academy of Sciences (UCAS)
[3]School of Electronic, Electrical and Communication Engineering, UCAS
[4]Key Laboratory of Target Cognition and Application Technology (TCAT) [5]Peng Cheng Lab
{huhuiyang22,wangpeijin17,bihanbo21,tongboyuan22,wangzhaozhi22}@mails.ucas.ac.cn

2025 IEEE/CVF International Conference on Computer Vision (ICCV) | 979-8-3315-8775-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCV51701.2025.00921

Abstract

Remote sensing foundation models largely break away from the traditional paradigm of designing task-specific models, offering greater scalability across multiple tasks. However, they face challenges such as low computational efficiency and limited interpretability, especially when dealing with large-scale remote sensing images. To overcome these, we draw inspiration from heat conduction, a physical process modeling local heat diffusion. Building on this idea, we are the first to explore the potential of using the parallel computing model of heat conduction to simulate the local region correlations in high-resolution remote sensing images, and introduce RS-vHeat, an efficient multi-modal remote sensing foundation model. Specifically, RS-vHeat 1) applies the Heat Conduction Operator (HCO) with a complexity of $O(N^{1.5})$ and a global receptive field, reducing computational overhead while capturing remote sensing object structure information to guide heat diffusion; 2) learns the frequency distribution representations of various scenes through a self-supervised strategy based on frequency domain hierarchical masking and multi-domain reconstruction; 3) significantly improves efficiency and performance over state-of-the-art techniques across 4 tasks and 10 datasets. Compared to attention-based remote sensing foundation models, we reduce memory usage by 84%, FLOPs by 24% and improves throughput by 2.7 times. The code will be made publicly available.

1. Introduction

Recently, remote sensing (RS) technology has become a vital data source for scientific research, resource management, and environmental monitoring [30, 47, 52, 76] by capturing surface information via satellites. Traditional models, designed as single-task networks for specific RS tasks [36, 50], struggle with multi-payload, multi-resolution, multi-temporal, and multi-feature RS data [20]. However, the emergence of remote sensing foundation models (RSFMs) has overcome these limitations, enabling unified handling of diverse tasks scenarios, significantly enhancing the scalability and versatility of the models [2, 3, 15, 31, 39–41, 44, 53, 56, 59, 60, 70]. RSFMs can automatically extract features from remote sensing imagery (RSI), providing a robust foundation for real-world tasks.

Prior RSFMs typically use visual encoders and decoders, leveraging large-scale RS datasets for pre-training while

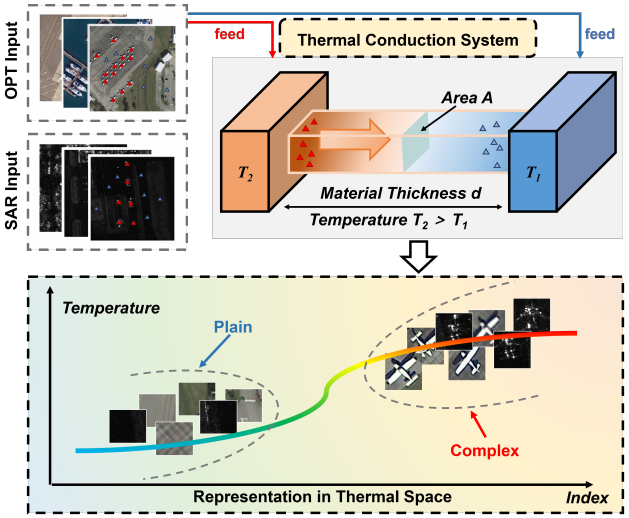


Figure 1. Calculation of RS images in the heat conduction system. In heat conduction theory, different materials exhibit varying diffusion rates. Inspired by this, the heat-conduction-based encoder maps optical (OPT) and synthetic aperture radar (SAR) into a unified thermal space, enhancing sensitivity to shape features by simulating heat flow and accumulation within the object region.

*Corresponding author

2380-7504/25/$31.00 ©2025 IEEE
DOI 10.1109/ICCV51701.2025.00921

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

incorporating self-supervised learning strategies. Specifically, RSFMs can be categorized based on the type of backbone used: CNN-based methods often use ResNet18/50 [25] to extract features [2, 31, 39, 40], while attention-based methods mainly use ViT [17] or Swin transformer [35] with attention mechanisms [55] to capture global dependencies [3, 15, 41, 44, 53, 56]. Both approaches use strategies such as masked reconstruction, knowledge distillation, or contrastive learning during pre-training to enhance model robustness (a detailed comparison is available in the supplementary materials). Despite the significant advancements achieved by these RSFMs, they still face two limitations:

Balancing Efficiency and Receptive Field. To accurately capture information about the large objects in RSI, model outputs must be responsive to sufficiently large regions [37]. However, this need significantly increases computational complexity [14, 38]. CNN-based networks lack a global receptive field due to their reliance on sliding computations with fixed-size convolutional kernels. While attention-based models achieve global modeling, their attention mechanisms incur quadratic computational complexity. Therefore, existing RSFMs struggle to deliver both fast and high-accuracy inference in practical applications.

Weak Physical Interpretability. RS objects often exhibit irregular polygonal shapes [29], and current RSFMs struggle to integrate physical principles to explain how object features propagate. This deficiency makes it challenging for researchers to adjust learning strategies effectively [43, 46, 54]. In the long term, RSFMs need to possess a certain degree of information interpretability.

To address these, this paper introduces RS-vHeat, a heat-conduction-based RSFM that supports multi-modal inputs inspired by the idea of vHeat [63]. **First**, heat conduction represents the natural process of energy diffusion from high- to low-temperature regions, as shown in the conceptual physical model in Fig. 1. This process transitions from an unsteady to a steady state based on the material. Since its computational process resembles feature extraction in neural networks, it can be applied to RS image processing. **Second**, we hypothesize that object types correspond to special feature distributions, with the model predicting diffusion rates based on RS-specific properties, simulating parameter computation through heat flow. This approach projects all modalities into a common thermal space, following the constraint that complex areas containing RS objects are high-temperature regions where heat accumulates, while sparse regions are low-temperature areas where heat diffuses easily, as shown in the lower part of Fig. 1. **Third**, building on this theory, we further design a RSFM with physical interpretability using 3 million optical and SAR data for pre-training, as shown in Fig. 2. The heat conduction network simulates the diffusion process of heat across large-scale multi-modal RS data, facilitating the alignment of feature propagation with the structural characteristics of the objects. Moreover, its computational approach provides guidance for the efficient operation of the network.

To summarize, our contributions are as follows:

1. We introduce RS-vHeat, a RSFM designed based on the heat conduction differential equation to process RS data. It conceptualizes the semantic relationships between pixels in RS images as the propagation of heat.
2. We propose a self-supervised strategy based on frequency domain hierarchical masking and multi-domain reconstruction that preserves small objects, driving the model to reconstruct fine and coarse frequency signals.
3. We design spatial correction embeddings, which operate directly on the global features to capture local details, assisting in simulating the rate of thermal diffusion.
4. We evaluate RS-vHeat on 10 datasets, showing it outperforms advanced RSFMs in accuracy while maintaining lower computational complexity. When processing large-scale images, RS-vHeat reduces memory usage by 84%, decreases FLOPs by 24%, and improves throughput by 2.7 times compared to attention-based RSFMs.

2. Related Work

2.1. Mainstream Network Architectures in RS

Convolutional Neural Networks. CNNs have been adapted for RS to tackle challenges in satellite imagery [28, 74]. However, local receptive fields, mainly limited by the kernel size, restrict the capture of long-range dependencies [17]. This drawback is especially critical in RS, where large-scale patterns and complex spatial relationships are prevalent, making the balance between local feature extraction and global context a key research focus [11, 16].

Transformers. Self-attention mechanisms [55] empower networks to capture long-range dependencies [17, 35]. Research studies have shown that plain ViTs performs better than traditional CNN models in RS applications [56, 72]. However, the limitation stems from the ViTs' restricted ability to effectively handle long sequences. When processing large-scale RS images, the computational load of networks grows quadratically, leading to significant overhead.

Mamba-Based Models. Mamba, an efficient implementation of State Space Models (SSMs) [23], leverages selective scan mechanisms for long-sequence processing [34, 78], balancing computational efficiency and high accuracy. [22, 71]. It has been rapidly adopted in RS [8, 9, 79], with its ability to process long sequences, although Mamba, as an innovative architecture, suffers from limited interpretability.

2.2. Self-Supervised Learning Strategies for RSFMs

Contrastive Learning. By establishing rules to distinguish between positive and negative samples, contrastive learning aims to fully understand the relationships between these

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

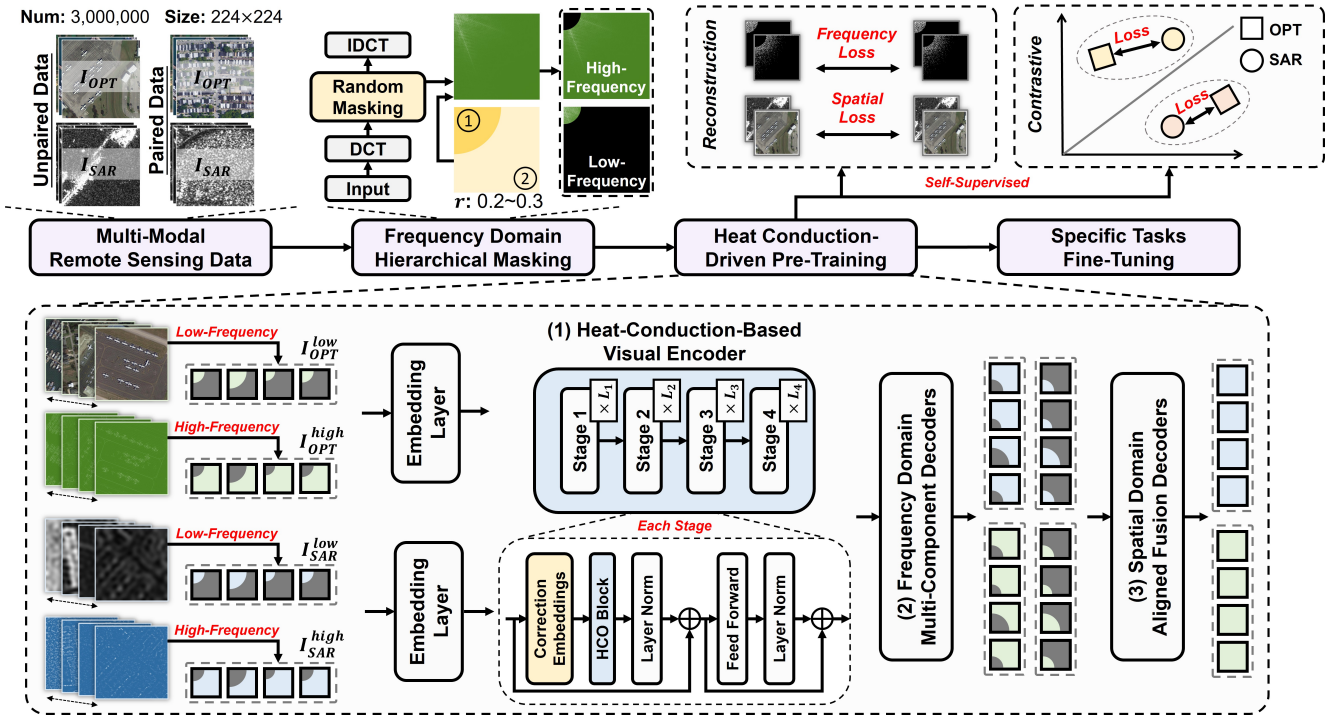


Figure 2. The pre-training process of RS-vHeat. It performs frequency domain hierarchical masking based on randomly generated sector regions to separate each image into high- and low-frequency components. These component images are fed into the network and projected into the thermal space via embedding layers. The heat diffusion is computed within the heat-conduction-based visual encoder to simulate complex RS objects. The deep features undergo multi-domain reconstruction loss and contrastive loss computation via decoders.

samples. GASSL [2] treats RS images of the same scene captured at different times as positive pairs, using them as self-supervised signals. Based on seasonal contrast method, SeCo [40] and CACo [39] effectively leverage temporal information within the networks by comparing scene variations across different years or seasons as perceptual signals. SkySense [24] utilizes multi-modal RS images as input and implements a multi-granularity contrastive learning framework. Motivated by the aforementioned methods, we apply contrastive constraints in the thermal space to optical and SAR data, encouraging the model to focus on deep, fine-grained semantic relationships within the RS images.

Masked Image Modeling. By involving masking parts of the image and predicting the missing information, networks can understand the details of RSI. RingMo [51] attempts to solve the issue that directly masking image patches can easily lead to the loss of small objects by designing an incomplete masking strategy that is implemented proportionally within the patches. SpectralGPT [26] models multi-spectral data as three-dimensional data and applies masking in three-dimensional space. Scale-MAE [44] masks pixels in the spatial domain and reconstructs high- and low-frequency images to learn image representations at different scales. Inspired by these, to preserve and learn the information of complex RS scenes, we employ dual reconstruction in both the spatial and frequency domains as constraints.

3. Proposed RS-vHeat

We introduce RS-vHeat, a multi-modal remote sensing foundation model with three key components: a frequency domain hierarchical masking strategy for multi-modal RS data, a visual encoder for modeling internal heat flow within RS images, and decoders for multi-domain reconstruction. As shown in Fig. 2, RS-vHeat enables shared representation learning of multi-modal data and projection into a thermal space to simulate heat propagation. Our architecture is inspired by vHeat [63], but differs in its emphasis on leveraging heat propagation within the self-supervised process, where we introduce a novel internal design. Details of vHeat are provided in the supplementary materials.

Frequency Domain Hierarchical Masking Strategy. Traditional spatial masking [69] on RS images often obscures smaller objects, hindering reconstruction. Inspired by the frequency-aware dynamic network [68], we implement a frequency domain hierarchical masking strategy to preserve object structure, enabling more accurate heat propagation by focusing on frequency domain signals.

Specifically, given the input (optical and SAR), denoted as $I(x, y, c) = \{I_o, I_s\}, I_o \in \mathbb{R}^{H\times W\times 3}, I_s \in \mathbb{R}^{H\times W\times 1}$, two parallel streams process paired and unpaired data. The Discrete Cosine Transform (DCT) is applied along each image dimension, converting $I(x, y)$ to its frequency

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

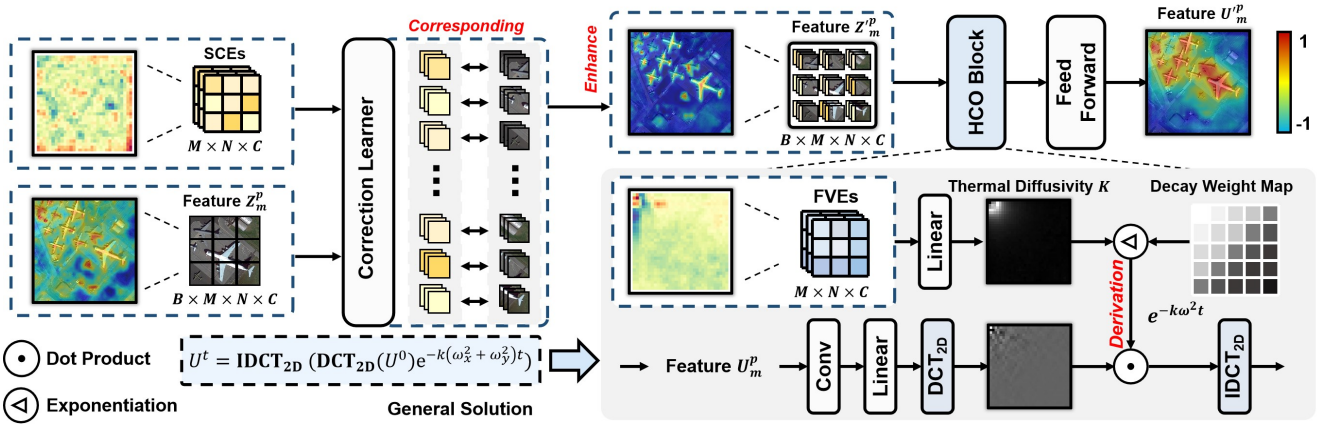


Figure 3. The overall structure of the heat-conduction-based visual encoder, simulates the general solution of heat conduction. SCEs first dynamically perform spatial domain correction, adjusting heat distribution based on the RS scene. The enhanced image undergoes a 2D DCT in the HCO block, interacting with the heat diffusion rate predicted by the FVEs to complete a heat conduction calculation, which is then transformed back into a visual representation via 2D IDCT.

representation $\tilde{I}(u,v)$, where low-frequency components are concentrated in the top-left of the spectrum. A sector mask separates high-frequency $\tilde{I}^{high}(u,v)$ and low-frequency $\tilde{I}^{low}(u,v)$ regions, using a random masking rate of 20%-30%. Next, the Inverse Discrete Cosine Transform (IDCT) is applied along the image dimensions to revert the data back to the spatial domain, obtaining $I^{low}(x,y)$ and $I^{high}(x,y)$. Notably, the DCT and IDCT operations are differentiable, enabling efficient computation and seamless integration into network training on both CPUs and GPUs. Details can be found in the supplementary material.

Heat-Conduction-Based Visual Encoder. High- and low-frequency information from multi-modal data is mapped to a shared thermal space and then fed into the heat-conduction-based visual encoder for thermal simulation. In most RS downstream tasks, areas containing RS objects tend to appear as high-temperature zones, while sparse or empty regions display as low-temperature zones.

The Heat Conduction Operator (HCO) simulates the process of visual information transmission as thermal conduction. The two-dimensional temperature distribution at time t, $u(x,y,t)$ is extended to a multi-dimensional feature distribution $U(x,y,c,t)$. The HCO block specifically models the general solution of physical heat conduction:

$$U_m^t = \mathcal{F}^{-1}\left(\mathcal{F}(U_m^0)e^{-k(\omega_x^2+\omega_y^2)t}\right) \tag{1}$$

where U^0 and U^t represents the input $U(x,y,c,0)$ and output $U(x,y,c,t)$. We denote the Discrete Fourier Transform (DFT) and its inverse (IDFT) as $\mathcal{F}$ and $\mathcal{F}^{-1}$. $m \in \{o,s\}$ refers to the modality (optical or SAR). Based on the Neumann boundary condition [12], we replace the 2D DFT and IDFT with the 2D DCT ($\mathrm{DCT_{2D}}$) and IDCT ($\mathrm{IDCT_{2D}}$) [48]:

$$U_m^t = IDCT_{2D}\left(DCT_{2D}(U_m^0)e^{-k(\omega_x^2+\omega_y^2)t}\right) \tag{2}$$

where $e^{-k(\omega_x^2+\omega_y^2)t}$ functions as an adaptive filter in the frequency domain, executing heat conduction.

A set of learnable frequency value embeddings (FVEs) is predicted to estimate the heat diffusion coefficient k ($k := k(\omega_x,\omega_y)$), facilitating adaptive heat transfer, as shown in Fig. 3. The weights are denoted as $W_{FVEs} \in \mathbb{R}^{M\times N\times C}$:

$$U_m'^p = HCO(U_m^p, W_{FVEs}) \tag{3}$$

where U and $U\prime$ represent the temperature states of the features from different modalities before and after passing through the HCO block, respectively, with $p \in \{high, low\}$ indicating the frequency level.

To enhance the spatial representation of different structural components, we introduce a lightweight correction learner. By predicting a set of spatial correction embeddings (SCEs) from large-scale pre-trained images, it interacts with the existing temperature field and perform activation, as shown in Fig. 3. The weight $W_{SCEs} \in \mathbb{R}^{M\times N\times C}$ adaptively adjusts the object boundaries in the spatial domain based on content by adding them to the original image $I(x,y)$. Given the upper-layer temperature feature Z_m^p, the correction learner can be expressed as CL:

$$Z_m'^p = CL(Z_m^p, W_{SCEs}) \tag{4}$$

Where $Z_m'^p$ represents the temperature features after adaptive adjustment.

Multi-Domain Reconstruction Decoders. Considering that the restoration of heat distribution is modality-specific, the optical and SAR decoders (Fig. 2 (2)), D_o^δ and D_s^δ concurrently operate on the encoded outputs ($F_o'^p$, $F_s'^p \in \mathbb{R}^{M\times N\times C}$). By employing convolution and pixel shuffling, decoders effectively upsample feature maps. The reconstruction outputs I' for both modalities—$I_o'^{high}$, $I_o'^{low}$ for optical, and $I_s'^{high}$, $I_s'^{low}$ for SAR—are as follows:

$$I_o'^p = D_o^\delta(F_o'^p), I_s'^p = D_o^\delta(F_s'^p) \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

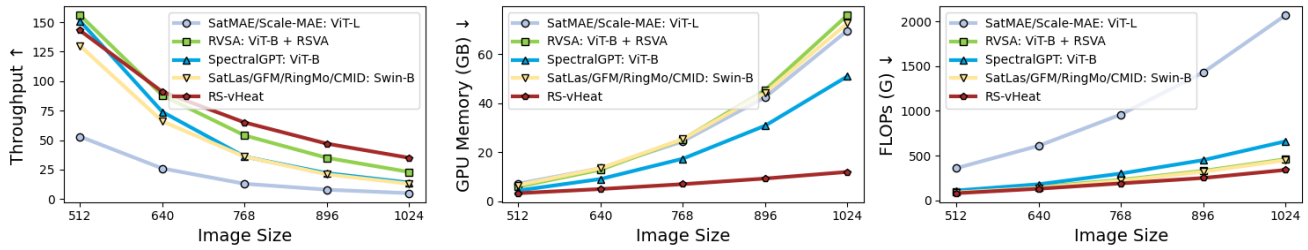


Figure 4. Comparison of throughput (left), GPU memory (middle), and FLOPs (right) across image sizes for RSFMs. RS-vHeat shows significantly higher throughput, lower memory usage and lower FLOPs than current attention-based RSFMs, especially for large-scale images. All tests were performed in a consistent environment using a single A100 (80G) GPU with a batch size of 32.

The reconstruction is guided by the $\mathcal{L}_1$ loss in the frequency domain. After applying DCT, the loss $\mathcal{L}_{Fre}$ is computed by measuring the difference between transformed features $\tilde{I}'^{p}_{m}(u,v)$ and $\tilde{I}^{p}_{m}(u,v)$.

To learn more robust higher-level features, spatial decoders (Fig. 2 (3)) are responsible for fusion and reconstruction. Specifically, outputs, F'^{p}_{m} and F^{p}_{m}, from the third and fourth stages extract modality-specific frequency features. The fusion layer integrates the frequency components into higher-level image features, F'_m and F_m:

$$\begin{aligned} F'_m &= Concat[CONV(F'^{high}_m), CONV(F'^{low}_m)] \\ F_m &= Concat[CONV(F^{high}_m), CONV(F^{low}_m)] \end{aligned} \tag{6}$$

Furthermore, the integration of multi-stage features is achieved through the function $P_m = g(F_m, F'_m)$, where g combines convolution, ReLU activation and element-wise summation operations to collectively produce the final high-level output $P_m = \{P_o, P_s\}$. Subsequently, these features are fed into modality-specific spatial domain decoders D^{ϕ}_{o} and D^{ϕ}_{s}, they are upsampled to the original image size to restore the high-level semantics of the original image:

$$I'_o = D^{\phi}_{o}(P_o), I'_s = D^{\phi}_{s}(P_s) \tag{7}$$

Where I'_o and I'_s represent the reconstructed high-level optical and SAR data. The reconstruction loss in the spatial domain $\mathcal{L}_{Spa}$ is then computed using the $\mathcal{L}_1$ loss.

To explore the network's understanding of fine-grained semantic information, we employ contrastive loss to compute the semantic discrepancy between different modalities. The loss operates on embeddings F^{p}_{m} obtained from two different preprocessing methods applied to the same image, representing low- and high-frequency features. The contrastive loss $\mathcal{L}_{Con}$ is calculated leveraging cosine similarity, following the computational methodology outlined in [1].

The overall loss function is shown in Eq. (8). The combined approach encourages the model to fully capture both global structures and fine-grained details:

$$\mathcal{L}_{total} = \mathcal{L}_{Con} + \mathcal{L}_{Spa} + \mathcal{L}_{Fre} \tag{8}$$

4. Experiments

4.1. Training Implementation

The training of RS-vHeat consists of self-supervised pre-training and downstream fine-tuning. The visual encoder follows the Swin-B [35] configuration with four stages of 2, 2, 18, and 2 blocks. During pre-training, we use a large-scale multi-modal dataset with 450k matched optical and SAR image pairs, totaling over 3 million entries, following the methodology of RingMo [51]. The model is trained on eight A100 (80G) GPUs for 200 epochs with images of size 224. The training starts with a learning rate of 1e-6 for 10 warm-up epochs, gradually increasing to 2e-4, with a cosine annealing schedule and a minimum learning rate of 1e-5. For fine-tuning, we transfer the pre-trained embedding layers and visual encoder structure, adjusting the fixed-size FVEs and SCEs to match the image dimensions.

4.2. Performance

We comprehensively evaluate the performance of RS-vHeat compared to other representative RSFMs. Primarily, as shown in Fig. 4, we analyze throughput, memory usage, and FLOPs across various image sizes. RS-vHeat (depicted by the red line) demonstrates significantly higher throughput, lower memory consumption and FLOPs, with these advantages becoming more pronounced as image sizes increase. For instance, when processing 1024×1024 resolution images, RS-vHeat achieves 2.7 times the throughput of Swin-B-based models such as SatLas [3], GFM [41], RingMo [51], and CMID [42], while reducing memory usage by 84%. Similarly, when compared to the improved ViT-B-based SpectralGPT [26], RS-vHeat achieves 2.5 times the throughput while reducing memory usage by 77%.

Next, we evaluate the accuracy and computational efficiency of fine-tuning RS-vHeat on four tasks—semantic segmentation, object detection, classification and change detection—to demonstrate the effectiveness, along with extensive ablation studies. Many RSFMs primarily focus on optical data and do not support SAR input, so we compare RS-vHeat with other RSFMs using optical datasets, while SAR-specific tasks are compared against specialized mod-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

Table 1. Comparison of mF1, mIoU, parameters, and FLOPs on iSAID and Potsdam with other RSFMs.

Backbone Type	Method	Backbone	Potsdam mF1 ↑	iSAID mIoU ↑	Image Size	Params	FLOPs ↓
CNN-Based	GASSL (ICCV'2021) [2]	ResNet-50	91.27	65.95	896^2	64M	722G
	SeCo (ICCV'2021) [40]	ResNet-50	89.03	57.20	896^2	64M	722G
	SSL4EO (GRSM'2023) [58]	ResNet-50	91.54	64.01	896^2	64M	722G
	CACo (CVPR'2023) [39]	ResNet-18	91.35	64.32	896^2	41M	**671G**
	TOV (JSTARS'2023) [53]	ResNet-50	92.03	66.24	896^2	64M	722G
	SAMRS (NeurIPS'2023) [57]	ResNet-50	91.43	66.26	896^2	64M	722G
	RS-vHeat (Ours)	vHeat-B + SCEs	**92.82**	**68.72**	896^2	148M	921G
ViT-Based	RVSA (TGRS'2022) [56]	ViT-B + RVSA	-	64.49	896^2	128M	1043G
	SatMAE (NeurIPS'2022)[15]	ViT-L	90.63	62.97	896^2	341M	1536G
	Scale-MAE (ICCV'2023) [44]	ViT-L	91.54	65.77	896^2	341M	1536G
	RS-vHeat (Ours)	vHeat-B + SCEs	**92.82**	**68.72**	896^2	148M	**921G**
Swin-Based	RingMo (TGRS'2022) [51]	Swin-B	91.27	67.20	896^2	121M	968G
	SatLas (ICCV'2023) [3]	Swin-B	91.28	68.71	896^2	121M	968G
	GFM (ICCV'2023) [41]	Swin-B	91.85	66.62	896^2	121M	968G
	CMID (TGRS'2023) [42]	Swin-B	91.86	66.21	896^2	121M	968G
	SkySense (CVPR'2024) [24]	Swin-H	**93.99**	**70.91**	896^2	>702M	>2708G
	RS-vHeat (Ours)	vHeat-B + SCEs	92.82	68.72	896^2	148M	**921G**

Table 2. Comparison of OA and user's accuracy on WHU-OPT-SAR with other specialized models.

Method	Publication	User's Accuracy ↑							OA ↑
		Farmland	City	Village	Water	Forest	Road	Others	
SegFormer [67]	NeurIPS'2021	79.1	72.9	38.0	64.7	88.1	0.3	0.4	75.5
Segmenter [49]	ICCV'2021	**82.3**	**75.2**	51.7	74.4	89.4	16.0	12.0	79.9
MCANet [32]	JAG'2022	74.3	62.2	53.1	65.7	**95.5**	31.0	9.8	82.9
VMamba [34]	NeurIPS'2024	82.1	74.1	57.8	**81.4**	89.1	41.5	18.4	81.4
MMOKD [33]	TGRS'2024	70.0	58.1	50.0	69.7	80.1	40.1	25.0	82.5
RS-vHeat (Ours)	-	81.1	67.3	**67.5**	79.0	90.2	**54.9**	**55.3**	**83.9**

Table 3. Comparison of mIoU, OA, and AA on AIR-PolSAR-Seg with other specialized models.

Method	Publication	mIoU ↑	OA ↑	AA ↑
DeepLab V3+ [10]	ECCV'2018	48.21	76.81	63.55
EncNet [75]	CVPR'2018	47.75	75.67	57.51
PSANet [77]	ECCV'2018	47.14	76.21	62.92
CCNet [27]	ICCV'2019	46.46	75.53	55.83
DANet [18]	CVPR'2019	51.93	76.91	62.79
GCNet [5]	ICCV'2019	47.56	76.75	57.10
RS-vHeat (Ours)	-	**57.46**	**81.46**	**65.92**

els. RSFMs are categorized into three main groups based on the backbone: CNN-based, ViT-based, and Swin-based. All datasets follow official partitioning methods for training and testing. More details on datasets and visualizations are provided in the supplementary materials.

Single- and Multi-Modal Semantic Segmentation. Evaluation is conducted on two optical datasets (Potsdam [47], iSAID [64]), one SAR dataset (Air-PolSAR-Seg [61]), and one multi-modal dataset (WHU-OPT-SAR [32]), using UPerNet [66] as the output head. Tab. 1 compares RS-vHeat with 14 other RSFMs on iSAID and Potsdam. RS-vHeat surpasses CNN-based models in accuracy and achieves lower FLOPs than ViT-based models, improving over Scale-MAE [44] by 1.28% and 2.95%. It also exhibits the lowest FLOPs among Swin-based models, and substantially reduces parameters compared to SkySense [24], with only 1.17% and 2.19% decreases in accuracy. Besides, on AIR-PolSAR-Seg (Tab. 3) and WHU-OPT-SAR (Tab. 2), RS-vHeat outperforms other specialized segmentation models, demonstrating the applicability of the heat conduction model in SAR and multi-modal segmentation tasks.

Object Detection. We conduct coarse- and fine-grained experiments on two optical datasets (FAIR1M [52], DIOR [30]) and one SAR dataset (SAR-AIRcraft-1.0 [62]), using YOLOX [21] as the output head. On the DIOR dataset, Tab. 4 shows that RS-vHeat demonstrates not only lower FLOPs but also surpasses the results of SkySense [24] by 3.57%. As shown in Tab. 6 and Tab. 5, RS-vHeat also achieves strong performance and high computational efficiency on two additional fine-grained datasets.

Change Detection. We train and test on the LEVIR-CD dataset [6], employing the BIT architecture [7] with cross-entropy loss. RS-vHeat demonstrates superior adaptability, achieving an F1 score of 93.48%, outperforming existing methods and surpassing SkySense [24] by 0.9 points, as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

Table 4. Comparison of mAP_{50}, parameters, and FLOPs on DIOR with other RSFMs.

Backbone Type	Method	Backbone	Image Size	mAP_{50}↑	Params	FLOPs ↓
CNN-Based	GASSL (ICCV'2021) [2]	ResNet-50	800^2	67.40	41M	134G
	CACo (CVPR'2023) [39]	ResNet-18	800^2	66.91	28M	**101G**
	TOV (JSTARS'2023) [53]	ResNet-50	800^2	70.16	41M	134G
	SSL4EO (GRSM'2023) [58]	ResNet-50	800^2	64.82	41M	134G
	RS-vHeat (Ours)	vHeat-B + SCEs	800^2	**82.30**	128M	266G
ViT-Based	RVSA (TGRS'2022) [56]	ViT-B + RVSA	800^2	73.22	113M	378G
	SatMAE (NeurIPS'2022)[15]	ViT-L	800^2	70.89	324M	1094G
	Scale-MAE (ICCV'2023) [44]	ViT-L	800^2	73.81	324M	1094G
	RS-vHeat (Ours)	vHeat-B + SCEs	800^2	**82.30**	128M	**266G**
Swin-Based	RingMo (TGRS'2022) [51]	Swin-B	800^2	75.90	105M	322G
	SatLas (ICCV'2023) [3]	Swin-B	800^2	74.10	105M	322G
	GFM (ICCV'2023) [41]	Swin-B	800^2	72.84	105M	322G
	CMID (TGRS'2023) [42]	Swin-B	800^2	75.11	105M	322G
	SkySense (CVPR'2024) [24]	Swin-H	800^2	78.73	>674M	>1679G
	RS-vHeat (Ours)	vHeat-B + SCEs	800^2	**82.30**	128M	**266G**

Table 5. Comparison of mAP, parameters, and FLOPs on FAIR1M-2.0 with other RSFMs.

Method	Backbone	mAP ↑	Params	FLOPs ↓
CACo [39]	ResNet-18	47.83	42M	**64G**
GASSL [2]	ResNet-50	48.15	55M	77G
TOV [53]	ResNet-50	49.62	55M	77G
SSL4EO [58]	ResNet-50	49.37	55M	77G
RVSA [56]	ViT-B + RVSA	47.04	126M	160G
SatMAE [15]	ViT-L	46.55	336M	241G
Scale-MAE [44]	ViT-L	48.31	336M	241G
SatLas [3]	Swin-B	46.19	119M	167G
GFM [41]	Swin-B	49.69	119M	167G
RingMo [51]	Swin-B	46.21	119M	167G
CMID [42]	Swin-B	50.58	119M	167G
SkySense [24]	Swin-H	**54.57**	>688M	>900G
RS-vHeat	vHeat-B + SCEs	48.29	130M	137G

Table 6. Comparison of mAP_{50} and mAP_{75} on SAR-AIRcraft-1.0 with other specialized models.

Method	Publication	mAP_{50} ↑	mAP_{75} ↑
Faster R-CNN [45]	TPAMI'2016	76.1	62.2
Cascade R-CNN [4]	CVPR'2018	75.7	58.9
RepPoints [73]	ICCV'2019	72.6	53.3
SKG-Net [19]	JSTARS'2021	70.7	46.4
SA-Net [62]	RADARS'2023	77.7	62.8
RS-vHeat (Ours)	-	**87.1**	**67.4**

shown in Tab. 7. Despite a 256-pixel input size, RS-vHeat maintains FLOPs comparable to the Swin-B-based baseline while delivering higher precision.

Image Classification. We validate our model on two benchmark datasets (AID [65], NWPU-RESISC45 [13]) by attaching a classification head designed, and employ cross-entropy loss for computation. As shown in Tab. 8, RS-vHeat surpasses CNN-based models like CACo [39] in classification accuracy. Compared to ViT- and Swin-based models,

Table 7. Comparison of F1 on LEVIR-CD with other RSFMs.

Method	Backbone	F1 ↑	Params	FLOPs ↓
CACo [39]	ResNet-18	81.04	12M	**11G**
GASSL [2]	ResNet-50	78.19	27M	25G
SeCo [40]	ResNet-50	90.14	27M	25G
SSL4EO [58]	ResNet-50	89.05	27M	25G
RVSA [56]	ViT-B + RVSA	90.86	94M	57G
SatMAE [15]	ViT-L	87.65	304M	162G
Scale-MAE [44]	ViT-L	92.07	304M	162G
SatLas [3]	Swin-B	90.62	88M	45G
GFM [41]	Swin-B	91.73	88M	45G
RingMo [51]	Swin-B	91.86	88M	45G
CMID [42]	Swin-B	91.72	88M	45G
SkySense [24]	Swin-H	92.58	>656M	>307G
RS-vHeat (Ours)	vHeat-B + SCEs	**93.48**	93M	46G

RS-vHeat demonstrates superior computational efficiency while also achieving competitive accuracy.

4.3. Ablations

To validate the effectiveness of the features learned through pre-training based on heat conduction theory, we conduct ablation studies on key components.

Effectiveness of Masked Training. We visualize the accuracy curves of the visual encoder before and after introducing new structures across downstream tasks, as shown in Fig. 5. All three networks are pre-trained on the same multi-modal RS data. In (a), we use vHeat-B with the SimMIM [69] pixel mask training scheme; in (b), we add frequency domain masking and multi-domain reconstruction; and in (c), we integrate the innovative RS-vHeat structure, which includes SCEs. For optical tasks, RS-vHeat shows faster accuracy improvement, stabilizing at a higher ceiling. It surpasses 80% mAP_{50} on the DIOR dataset by the 120th epoch. For SAR tasks, while there are fluctuations, RS-vHeat demonstrates better adaptability and prediction accuracy. Overall, the improvements in pre-training and SCEs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

Table 8. Comparison of OA, parameters, and FLOPs on AID and NWPU-RESISC45 with other RSFMs.

Backbone Type	Method	Backbone	AID		NWPU-RESISC45		Image Size	Params	FLOPs ↓
			TR=20%	TR=50%	TR=10%	TR=20%			
CNN-Based	GASSL (ICCV'2021) [2]	ResNet-50	93.55	95.92	90.86	93.06	1024^2	24M	87G
	SeCo (ICCV'2021) [40]	ResNet-50	93.47	95.99	89.64	92.91	1024^2	24M	87G
	CACo (CVPR'2023) [39]	ResNet-18	90.88	95.05	88.28	91.94	1024^2	11M	**38G**
	TOV (JSTARS'2023) [53]	ResNet-50	95.16	97.09	90.97	93.79	1024^2	24M	87G
	SSL4EO (GRSM'2023) [58]	ResNet-50	91.06	94.74	87.60	91.27	1024^2	24M	87G
	RS-vHeat (Ours)	vHeat-B + SCEs	**96.81**	**97.58**	**92.01**	**95.66**	1024^2	150M	340G
ViT-Based	RVSA (TGRS'2022) [56]	ViT-B + RVSA	**97.03**	**98.50**	**93.93**	**95.69**	1024^2	89M	460G
	SatMAE (NeurIPS'2022)[15]	ViT-L	95.02	96.94	91.72	94.10	1024^2	310M	2070G
	Scale-MAE (ICCV'2023) [44]	ViT-L	96.44	97.58	92.63	95.04	1024^2	310M	2070G
	RS-vHeat (Ours)	vHeat-B + SCEs	96.81	97.58	92.01	95.66	1024^2	150M	**340G**
Swin-Based	RingMo (TGRS'2022) [51]	Swin-B	96.90	98.34	94.25	95.67	1024^2	87M	450G
	SatLas (ICCV'2023) [3]	Swin-B	94.96	97.38	92.16	94.70	1024^2	87M	450G
	GFM (ICCV'2023) [41]	Swin-B	95.47	97.09	92.73	94.64	1024^2	87M	450G
	CMID (TGRS'2023) [42]	Swin-B	96.11	97.79	94.05	95.53	1024^2	87M	450G
	SkySense (CVPR'2024) [24]	Swin-H	**97.68**	**98.60**	**94.85**	**96.32**	1024^2	>660M	>2760G
	RS-vHeat (Ours)	vHeat-B + SCEs	96.81	97.58	92.01	95.66	1024^2	150M	**340G**

Table 9. Results of RS-vHeat under various constraints of self-supervised learning strategies.

#	Loss			Object Detection		Semantic Segmentation		
				DIOR (Optical)	SAR-AIRcraft-1.0 (SAR)	iSAID (Optical)	AIR-PolSAR-Seg (SAR)	WHU-OPT-SAR (Optical+SAR)
	SDR	FDR	CL	mAP_{50} ↑	mAP_{50} ↑	mIoU ↑	mIoU ↑	OA ↑
(a)	✓	✗	✗	78.2	84.6	66.1	54.7	80.5
(b)	✓	✓	✗	79.5	86.4	67.2	55.1	82.3
(c)	✓	✓	✓	**82.3**	**87.1**	**68.7**	**57.5**	**83.9**

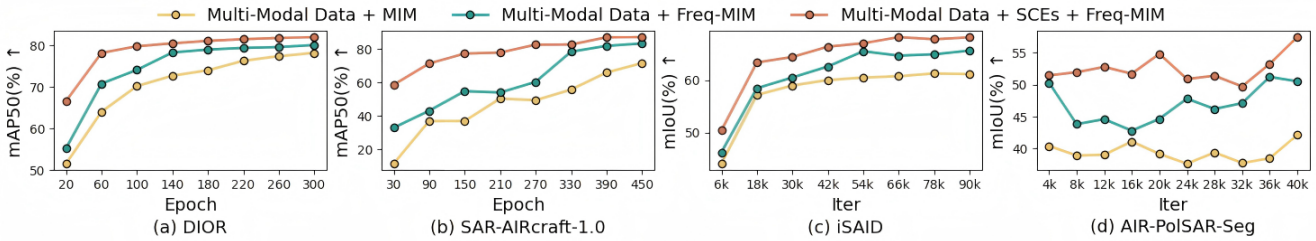


Figure 5. Accuracy curves of various methods across modalities. (a) optical object detection on DIOR, (b) SAR object detection on SAR-AIRcraft-1.0, (c) optical semantic segmentation on iSAID, and (d) SAR semantic segmentation on AIR-PolSAR-Seg.

lead to advanced feature extraction performance.

Reconstruction Learning Strategy. Tab. 9 demonstrates the effectiveness of different loss components—spatial domain reconstruction loss (SDR), frequency domain reconstruction loss (FDR), and contrastive loss (CL). When only SDR is applied (row a), performance drops slightly on both optical and SAR datasets. Adding FDR (row b) improves performance by leveraging frequency domain information. Introducing CL (row c) further enhances performance by enforcing similarity constraints in the multi-modal feature space, improving feature learning. This leads to an 83.9% OA on WHU-OPT-SAR, surpassing rows (a) and (b) by 3.4% and 1.6%, respectively. These results highlight the importance of combining these three losses for improved predictions across tasks and modalities.

5. Conclusion

In this work, we introduce the concept of heat conduction into RS tasks for the first time and establish a multi-modal RSFM, RS-vHeat. By employing a self-supervised learning strategy that integrates frequency domain masking and multi-domain reconstruction, along with a heat conduction operator incorporating spatial correction embeddings, we propose an approach that balances computational complexity and global receptive field coverage in RS. Our method captures global details from both spatial and frequency domains, significantly reducing the issue of small object omission, and achieves consistent multi-modal feature representation by mapping images into a high-dimensional thermal space. In future work, we plan to propose new solutions that offer fresh insights and methods for tackling visual modeling challenges across various industries.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

6. Acknowledgment

This work is supported by the National Natural Science Foundation of China under Grant 62331027, and the Key Laboratory of Target Cognition and Application Technology under Grant 2023-CXPT-LC-005.

References

[1] Sara Atito, Muhammad Awais, and Josef Kittler. Sit: Self-supervised vision transformer. *arXiv preprint arXiv:2104.03602*, 2021. 5

[2] Kumar Ayush, Burak Uzkent, Chenlin Meng, Kumar Tanmay, Marshall Burke, David Lobell, and Stefano Ermon. Geography-aware self-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10181–10190, 2021. 1, 2, 3, 6, 7, 8

[3] Favyen Bastani, Piper Wolters, Ritwik Gupta, Joe Ferdinando, and Aniruddha Kembhavi. Satlaspretrain: A large-scale dataset for remote sensing image understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16772–16782, 2023. 1, 2, 5, 6, 7, 8

[4] Zhaowei Cai and Nuno Vasconcelos. Cascade r-cnn: Delving into high quality object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6154–6162, 2018. 7

[5] Yue Cao, Jiarui Xu, Stephen Lin, Fangyun Wei, and Han Hu. Gcnet: Non-local networks meet squeeze-excitation networks and beyond. In *Proceedings of the IEEE/CVF international conference on computer vision workshops*, pages 0–0, 2019. 6

[6] Hao Chen and Zhenwei Shi. A spatial-temporal attention-based method and a new dataset for remote sensing image change detection. *Remote Sensing*, 12(10), 2020. 6

[7] Hao Chen, Zipeng Qi, and Zhenwei Shi. Remote sensing image change detection with transformers. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–14, 2021. 6

[8] Hongruixuan Chen, Jian Song, Chengxi Han, Junshi Xia, and Naoto Yokoya. Changemamba: Remote sensing change detection with spatio-temporal state space model. *arXiv preprint arXiv:2404.03425*, 2024. 2

[9] Keyan Chen, Bowen Chen, Chenyang Liu, Wenyuan Li, Zhengxia Zou, and Zhenwei Shi. Rsmamba: Remote sensing image classification with state space model. *IEEE Geoscience and Remote Sensing Letters*, 2024. 2

[10] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *Proceedings of the European conference on computer vision (ECCV)*, pages 801–818, 2018. 6

[11] Xi Chen, Zhiqiang Li, Jie Jiang, Zhen Han, Shiyi Deng, Zhihong Li, Tao Fang, Hong Huo, Qingli Li, and Min Liu. Adaptive effective receptive field convolution for semantic segmentation of vhr remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 59(4):3532–3546, 2020. 2

[12] Alexander H-D Cheng and Daisy T Cheng. Heritage and early history of the boundary element method. *Engineering analysis with boundary elements*, 29(3):268–302, 2005. 4

[13] Gong Cheng, Junwei Han, and Xiaoqiang Lu. Remote sensing image scene classification: Benchmark and state of the art. *Proceedings of the IEEE*, 105(10):1865–1883, 2017. 7

[14] Emmanuel Christophe, Julien Michel, and Jordi Inglada. Remote sensing processing: From multicore to gpu. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 4(3):643–652, 2011. 2

[15] Yezhen Cong, Samar Khanna, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David Lobell, and Stefano Ermon. Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery. *Advances in Neural Information Processing Systems*, 35:197–211, 2022. 1, 2, 6, 7, 8

[16] Xiaohu Dong, Ruigang Fu, Yinghui Gao, Yao Qin, Yuanxin Ye, and Biao Li. Remote sensing object detection based on receptive field expansion block. *IEEE Geoscience and Remote Sensing Letters*, 19:1–5, 2021. 2

[17] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 2

[18] Jun Fu, Jing Liu, Haijie Tian, Yong Li, Yongjun Bao, Zhiwei Fang, and Hanqing Lu. Dual attention network for scene segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3146–3154, 2019. 6

[19] Kun Fu, Jiamei Fu, Zhirui Wang, and Xian Sun. Scattering-keypoint-guided network for oriented ship detection in high-resolution and large-scale sar images. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14:11162–11178, 2021. 7

[20] Kun Fu, Xian Sun, Xiaolan Qiu, Wenhui Diao, Zhiyuan Yan, Lijia Huang, and Hongfeng Yu. Multi-satellite integrated processing and analysis method under remote sensing big data. *National Remote Sensing Bulletin*, 25(3):691–707, 2021. 1

[21] Zheng Ge, Songtao Liu, Feng Wang, Zeming Li, and Jian Sun. Yolox: Exceeding yolo series in 2021. *arXiv preprint arXiv:2107.08430*, 2021. 6

[22] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023. 2

[23] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. *arXiv preprint arXiv:2111.00396*, 2021. 2

[24] Xin Guo, Jiangwei Lao, Bo Dang, Yingying Zhang, Lei Yu, Lixiang Ru, Liheng Zhong, Ziyuan Huang, Kang Wu, Dingxiang Hu, et al. Skysense: A multi-modal remote sensing foundation model towards universal interpretation for earth observation imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27672–27683, 2024. 3, 6, 7, 8

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

[25] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 2

[26] Danfeng Hong, Bing Zhang, Xuyang Li, Yuxuan Li, Chenyu Li, Jing Yao, Naoto Yokoya, Hao Li, Pedram Ghamisi, Xiuping Jia, et al. Spectralgpt: Spectral remote sensing foundation model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024. 3, 5

[27] Zilong Huang, Xinggang Wang, Lichao Huang, Chang Huang, Yunchao Wei, and Wenyu Liu. Ccnet: Criss-cross attention for semantic segmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 603–612, 2019. 6

[28] Haifeng Li, Haikuo Huang, Li Chen, Jian Peng, Haozhe Huang, Zhenqi Cui, Xiaoming Mei, and Guohua Wu. Adversarial examples for cnn-based sar image classification: An experience study. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14:1333–1347, 2020. 2

[29] Jiang Li and Ram M Narayanan. A shape-based approach to change detection of lakes using time series remote sensing images. *IEEE transactions on geoscience and remote sensing*, 41(11):2466–2477, 2003. 2

[30] Ke Li, Gang Wan, Gong Cheng, Liqiu Meng, and Junwei Han. Object detection in optical remote sensing images: A survey and a new benchmark. *ISPRS journal of photogrammetry and remote sensing*, 159:296–307, 2020. 1, 6

[31] Wenyuan Li, Keyan Chen, Hao Chen, and Zhenwei Shi. Geographical knowledge-driven representation learning for remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–16, 2021. 1, 2

[32] Xue Li, Guo Zhang, Hao Cui, Shasha Hou, Shunyao Wang, Xin Li, Yujia Chen, Zhijiang Li, and Li Zhang. Mcanet: A joint semantic segmentation framework of optical and sar images for land use classification. *International Journal of Applied Earth Observation and Geoinformation*, 106: 102638, 2022. 6

[33] Xiao Liu, Fei Jin, Shuxiang Wang, Jie Rui, Xibing Zuo, Xiaobing Yang, and Chuanxiang Cheng. Multimodal online knowledge distillation framework for land use/cover classification using full or missing modalities. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 6

[34] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, and Yunfan Liu. Vmamba: Visual state space model, 2024. 2, 6

[35] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021. 2, 5

[36] Xiaonan Lu, Xian Sun, Wenhui Diao, Yingchao Feng, Peijin Wang, and Kun Fu. Lil: Lightweight incremental learning approach through feature transfer for remote sensing image scene classification. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–20, 2021. 1

[37] Wenjie Luo, Yujia Li, Raquel Urtasun, and Richard Zemel. Understanding the effective receptive field in deep convolutional neural networks. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 2016. 2

[38] Yan Ma, Haiping Wu, Lizhe Wang, Bormin Huang, Rajiv Ranjan, Albert Zomaya, and Wei Jie. Remote sensing big data computing: Challenges and opportunities. *Future Generation Computer Systems*, 51:47–60, 2015. Special Section: A Note on New Trends in Data-Aware Scheduling and Resource Provisioning in Modern HPC Systems. 2

[39] Utkarsh Mall, Bharath Hariharan, and Kavita Bala. Change-aware sampling and contrastive learning for satellite images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5261–5270, 2023. 1, 2, 3, 6, 7, 8

[40] Oscar Manas, Alexandre Lacoste, Xavier Giró-i Nieto, David Vazquez, and Pau Rodriguez. Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9414–9423, 2021. 2, 3, 6, 7, 8

[41] Matías Mendieta, Boran Han, Xingjian Shi, Yi Zhu, and Chen Chen. Towards geospatial foundation models via continual pretraining. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16806–16816, 2023. 1, 2, 5, 6, 7, 8

[42] Dilxat Muhtar, Xueliang Zhang, Pengfeng Xiao, Zhenshi Li, and Feng Gu. Cmid: A unified self-supervised learning framework for remote sensing image understanding. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–17, 2023. 5, 6, 7, 8

[43] Adrián Pérez-Suay, Jose E. Adsuara, María Piles, Laura Martínez-Ferrer, Emiliano Díaz, Alvaro Moreno-Martínez, and Gustau Camps-Valls. Interpretability of recurrent neural networks in remote sensing. In *IGARSS 2020 - 2020 IEEE International Geoscience and Remote Sensing Symposium*, pages 3991–3994, 2020. 2

[44] Colorado J Reed, Ritwik Gupta, Shufan Li, Sarah Brockman, Christopher Funk, Brian Clipp, Kurt Keutzer, Salvatore Candido, Matt Uyttendaele, and Trevor Darrell. Scale-mae: A scale-aware masked autoencoder for multiscale geospatial representation learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4088–4099, 2023. 1, 2, 3, 6, 7, 8

[45] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, 39(6):1137–1149, 2016. 7

[46] Huanfeng Shen, Menghui Jiang, Jie Li, Chenxia Zhou, Qiangqiang Yuan, and Liangpei Zhang. Coupling model- and data-driven methods for remote sensing image restoration and fusion: Improving physical interpretability. *IEEE Geoscience and Remote Sensing Magazine*, 10(2):231–249, 2022. 2

[47] Jamie Sherrah. Fully convolutional networks for dense semantic labelling of high-resolution aerial imagery. *arXiv preprint arXiv:1606.02585*, 2016. 1, 6

[48] Gilbert Strang. The discrete cosine transform. *SIAM review*, 41(1):135–147, 1999. 4

[49] Robin Strudel, Ricardo Garcia, Ivan Laptev, and Cordelia Schmid. Segmenter: Transformer for semantic segmenta-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

tion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 7262–7272, 2021. 6

[50] Xian Sun, Peijin Wang, Cheng Wang, Yingfei Liu, and Kun Fu. Pbnet: Part-based convolutional neural network for complex composite object detection in remote sensing imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 173:50–65, 2021. 1

[51] Xian Sun, Peijin Wang, Wanxuan Lu, Zicong Zhu, Xiaonan Lu, Qibin He, Junxi Li, Xuee Rong, Zhujun Yang, Hao Chang, et al. Ringmo: A remote sensing foundation model with masked image modeling. *IEEE Transactions on Geoscience and Remote Sensing*, 2022. 3, 5, 6, 7, 8

[52] Xian Sun, Peijin Wang, Zhiyuan Yan, Feng Xu, Ruiping Wang, Wenhui Diao, Jin Chen, Jihao Li, Yingchao Feng, Tao Xu, et al. Fair1m: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 184:116–130, 2022. 1, 6

[53] Chao Tao, Ji Qi, Guo Zhang, Qing Zhu, Weipeng Lu, and Haifeng Li. Tov: The original vision model for optical remote sensing image understanding via self-supervised learning. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16:4916–4930, 2023. 1, 2, 6, 7, 8

[54] Anastasios Temenos, Nikos Temenos, Maria Kaselimi, Anastasios Doulamis, and Nikolaos Doulamis. Interpretable deep learning framework for land use and land cover classification in remote sensing using shap. *IEEE Geoscience and Remote Sensing Letters*, 20:1–5, 2023. 2

[55] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. 2

[56] Di Wang, Qiming Zhang, Yufei Xu, Jing Zhang, Bo Du, Dacheng Tao, and Liangpei Zhang. Advancing plain vision transformer toward remote sensing foundation model. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–15, 2022. 1, 2, 6, 7, 8

[57] Di Wang, Jing Zhang, Bo Du, Minqiang Xu, Lin Liu, Dacheng Tao, and Liangpei Zhang. Samrs: Scaling-up remote sensing segmentation dataset with segment anything model. *Advances in Neural Information Processing Systems*, 36, 2024. 6

[58] Yi Wang, Nassim Ait Ali Braham, Zhitong Xiong, Chenying Liu, Conrad M Albrecht, and Xiao Xiang Zhu. Ssl4eo-s12: A large-scale multimodal, multitemporal dataset for self-supervised learning in earth observation [software and data sets]. *IEEE Geoscience and Remote Sensing Magazine*, 11(3):98–106, 2023. 6, 7, 8

[59] Yi Wang, Conrad M Albrecht, and Xiao Xiang Zhu. Multilabel guided soft contrastive learning for efficient earth observation pretraining. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 1

[60] Yi Wang, Conrad M. Albrecht, Nassim Ait Ali Braham, Chenying Liu, Zhitong Xiong, and Xiao Xiang Zhu. Decoupling common and unique representations for multimodal self-supervised learning. In *Computer Vision – ECCV 2024*, pages 286–303, Cham, 2025. Springer Nature Switzerland. 1

[61] Zhirui Wang, Xuan Zeng, Zhiyuan Yan, Jian Kang, and Xian Sun. Air-polsar-seg: A large-scale data set for terrain segmentation in complex-scene polsar images. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15:3830–3841, 2022. 6

[62] Zhirui Wang, Yuzhuo Kang, Xuan Zeng, Yuelei Wang, Ting Zhang, and Xian Sun. Sar-aircraft-1.0: High-resolution sar aircraft detection and recognition dataset. *Journal of Radars*, 12(4):906–922, 2023. 6, 7

[63] Zhaozhi Wang, Yue Liu, Yunfan Liu, Hongtian Yu, Yaowei Wang, Qixiang Ye, and Yunjie Tian. vheat: Building vision models upon heat conduction. *arXiv preprint arXiv:2405.16555*, 2024. 2, 3

[64] Syed Waqas Zamir, Aditya Arora, Akshita Gupta, Salman Khan, Guolei Sun, Fahad Shahbaz Khan, Fan Zhu, Ling Shao, Gui-Song Xia, and Xiang Bai. isaid: A large-scale dataset for instance segmentation in aerial images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 28–37, 2019. 6

[65] Gui-Song Xia, Jingwen Hu, Fan Hu, Baoguang Shi, Xiang Bai, Yanfei Zhong, Liangpei Zhang, and Xiaoqiang Lu. Aid: A benchmark data set for performance evaluation of aerial scene classification. *IEEE Transactions on Geoscience and Remote Sensing*, 55(7):3965–3981, 2017. 7

[66] Tete Xiao, Yingcheng Liu, Bolei Zhou, Yuning Jiang, and Jian Sun. Unified perceptual parsing for scene understanding. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018. 6

[67] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. *Advances in neural information processing systems*, 34: 12077–12090, 2021. 6

[68] Wenbin Xie, Dehua Song, Chang Xu, Chunjing Xu, Hui Zhang, and Yunhe Wang. Learning frequency-aware dynamic network for efficient super-resolution. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4308–4317, 2021. 3

[69] Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A simple framework for masked image modeling. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9653–9663, 2022. 3, 7

[70] Zhitong Xiong, Yi Wang, Fahong Zhang, Adam J. Stewart, Joëlle Hanna, Damian Borth, Ioannis Papoutsis, Bertrand Le Saux, Gustau Camps-Valls, and Xiao Xiang Zhu. Neural plasticity-inspired multimodal foundation model for earth observation, 2024. 1

[71] Rui Xu, Shu Yang, Yihui Wang, Bo Du, and Hao Chen. A survey on vision mamba: Models, applications and challenges. *arXiv preprint arXiv:2404.18861*, 2024. 2

[72] Tianyu Yan, Zifu Wan, and Pingping Zhang. Fully transformer network for change detection of remote sensing images. In *Proceedings of the Asian Conference on Computer Vision*, pages 1691–1708, 2022. 2

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.

[73] Ze Yang, Shaohui Liu, Han Hu, Liwei Wang, and Stephen Lin. Reppoints: Point set representation for object detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9657–9666, 2019. 7

[74] Chi Zhang, Shiqing Wei, Shunping Ji, and Meng Lu. Detecting large-scale urban land cover changes from very high resolution remote sensing images using cnn-based classification. *ISPRS International Journal of Geo-Information*, 8(4): 189, 2019. 2

[75] Hang Zhang, Kristin Dana, Jianping Shi, Zhongyue Zhang, Xiaogang Wang, Ambrish Tyagi, and Amit Agrawal. Context encoding for semantic segmentation. In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, pages 7151–7160, 2018. 6

[76] Yuxiang Zhang, Wei Li, Ran Tao, Jiangtao Peng, Qian Du, and Zhaoquan Cai. Cross-scene hyperspectral image classification with discriminative cooperative alignment. *IEEE Transactions on Geoscience and Remote Sensing*, 59(11): 9646–9660, 2021. 1

[77] Hengshuang Zhao, Yi Zhang, Shu Liu, Jianping Shi, Chen Change Loy, Dahua Lin, and Jiaya Jia. Psanet: Point-wise spatial attention network for scene parsing. In *Proceedings of the European conference on computer vision (ECCV)*, pages 267–283, 2018. 6

[78] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang. Vision mamba: Efficient visual representation learning with bidirectional state space model. *arXiv preprint arXiv:2401.09417*, 2024. 2

[79] Qinfeng Zhu, Yuanzhi Cai, Yuan Fang, Yihan Yang, Cheng Chen, Lei Fan, and Anh Nguyen. Samba: Semantic segmentation of remotely sensed images with state space model. *arXiv preprint arXiv:2404.01705*, 2024. 2

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:22:20 UTC from IEEE Xplore. Restrictions apply.